



Research paper

A multi-scale and horizon-adaptive Transformer for operational metocean forecasting in Korean coastal buoy networks

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ABSTRACT

Short-term multivariate metocean forecasts support coastal navigation, offshore operations, and weather-sensitive marine decisions. This study addresses multi-step forecasting from the Korean coastal buoy network using MetoFormer-HQ, a Transformer-based framework built on PatchTST with multi-scale patch embedding, gated channel interaction, and horizon-adaptive decoding with randomized-horizon training. On the Korean buoy benchmark, MetoFormer-HQ reduces root mean squared error (RMSE) and mean absolute error (MAE) by 5.74% and 10.11%, respectively, compared with the original PatchTST baseline (BASE), and by 30.02% and 37.24% compared with N-HiTS. To test whether forecasts merely track delayed observations, we add persistence, autoregressive, lag-sensitive, and event-window diagnostics. These analyses show that short-horizon wave-height prediction is strongly persistence-dominated, but MetoFormer-HQ improves zero-lag alignment and peak-window matching relative to BASE. The results support horizon-aware multivariate forecasting for Korean coastal buoy operations while clarifying the need for external forcing and spatial information for broader generalization.

1. Introduction

Accurate forecasting of marine environmental conditions is essential for a wide range of operational and public-safety tasks, including offshore renewable energy planning, ship routing and navigation safety, and coastal hazard mitigation. In particular, significant wave height (SWH), wind speed, current speed, and directional variables (e.g., wind direction and current direction) directly affect operational decisions and risk assessment in coastal waters. However, metocean time series are challenging due to concurrent periodicities (e.g., tidal and diurnal cycles), pronounced non-stationarity driven by weather systems, and complex nonlinear interactions among variables. Traditionally, metocean forecasting has relied on numerical prediction models, which are grounded in governing equations and can provide physically interpretable outputs, but their operational use is often constrained by computational cost, sensitivity to initial conditions, and the difficulty of providing sufficiently accurate high-resolution forecasts for localized regions (The WAMDI Group, 1988; Tolman et al., 2002; Ris et al., 1999; Hasselmann et al., 1973). As a complementary route, data-driven forecasting has gained momentum with the increasing availability of buoy observations and the rapid progress of deep learning.

Within Ocean Engineering, deep learning has been actively explored for short- to medium-range wave forecasting, particularly for

SWH, owing to its relevance to navigation safety, maritime operations, and coastal-energy decision support (De Girolamo et al., 2017; Zis et al., 2020; López-Ruiz et al., 2016). Earlier studies mainly examined recurrent or hybrid architectures that combine wind information with historical wave measurements for short lead times (Fan et al., 2020; Gao et al., 2021; Pushpam and Enigo, 2020). Subsequent work extended this line in several directions, including bathymetric augmentation for longer lead times (Jörges et al., 2021), transfer learning for data-scarce locations (Obara and Nakamura, 2022), and decomposition-based predictors (Zhao et al., 2023; Bekiryazıcı et al., 2025; S. Zhou et al., 2021; Ding et al., 2024). Even with these encouraging results, many wave-forecasting studies still focus on a limited target set, often SWH alone, are evaluated at fixed lead times, and do not explicitly account for the physical coupling among waves, winds, and currents (Han et al., 2022; Feng et al., 2022; Bai et al., 2022; Gracia et al., 2021; Minuzzi and Farina, 2023). More recent studies have improved particular prediction settings, yet analysis of multivariate interaction control and long-horizon error growth remains limited in unified multi-step forecasting settings (Domala et al., 2022; Zhang et al., 2024; Zheng et al., 2023; Huang and Dong, 2021; Shadmani et al., 2023). Taken together, these limitations point to the need for

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a forecasting framework that captures multi-scale temporal structure, preserves controlled cross-variable coupling, and adapts to the range of prediction horizons required in practice.

Recent advances in long-term time-series forecasting (LTSF) have shown competitive performance with Transformer-based architectures, and patch-based designs such as PatchTST provide an effective trade-off between accuracy and computational efficiency by representing time series as patch tokens (Nie et al., 2023). Nevertheless, applying a generic LTSF backbone to metocean forecasting faces several structural challenges. Metocean observations contain coexisting temporal patterns across multiple scales, ranging from short-term variability to tidal/diurnal cycles and longer synoptic/seasonal components, so a single-resolution tokenization may be insufficient to capture such concurrent periodicities (Wu et al., 2023; Challu et al., 2023). In addition, although channel-independent modeling can improve generalization in some settings (Nie et al., 2023), metocean variables are often physically coupled; fully independent processing can miss informative coupling, whereas indiscriminate mixing may amplify noise, which motivates controlled channel interaction (Zhang et al., 2023; Hu et al., 2018). Operational forecasting also frequently requires flexible lead times rather than a single fixed horizon, and training a model strictly for a fixed prediction length can limit transferability and practical usability when the demanded horizon changes, especially under regime shifts and extreme events.

Against this background, this study frames multivariate metocean forecasting as an operationally oriented Ocean Engineering problem rather than as a generic long-term time-series forecasting task. In practical coastal environments, forecasting systems must jointly represent the coupled behavior of waves, winds, currents, and related marine variables, while remaining reliable across multiple lead times that support navigation safety, offshore activities, and weather-sensitive marine decision making. To address these requirements, we develop *MetoFormer-HQ*, a horizon-aware forecasting framework built on PatchTST and tailored to coastal buoy observations. The proposed design combines multi-scale patch embedding, gated channel interaction, and horizon-adaptive decoding to better capture temporal periodicity, inter-variable dependence, and horizon-specific uncertainty. Through extensive experiments on the Korean coastal buoy benchmark, we show that the proposed framework provides a more practical and robust basis for short-term metocean forecasting in coastal marine applications.

The main contributions are summarized as follows:

- We formulate a unified multivariate and multi-horizon coastal-buoy forecasting problem that jointly predicts waves, winds, currents, temperatures, pressure, and directional encodings under a spatial out-of-distribution protocol.
- We introduce a PatchTST-based architecture that integrates multi-scale patch embedding, controlled channel interaction, and horizon-adaptive query decoding within a single operational forecasting framework.
- Beyond aggregate RMSE/MAE/ R^2 , we add persistence, autoregressive, lag-sensitive, and representative time-history diagnostics to directly assess whether the proposed forecasts contain useful lead-time information rather than only delayed tracking of recent observations.

2. Related work

2.1. Numerical wave models and data-driven motivation

Operational wave forecasting has traditionally relied on third-generation spectral wave models, including WAVEWATCH III, SWAN, and WAM, which provide physically interpretable predictions by representing wind input, nonlinear interactions, and dissipation processes (Tolman, 2014; Tolman et al., 2002; Booij et al., 1999; Ris et al., 1999; Zijlema, 2010; The WAMDI Group, 1988). For localized and rapidly

updated coastal operations, however, these models can be limited by high-resolution computational cost, sensitivity to forcing and boundary conditions, and site-specific calibration requirements (Amarouche et al., 2021; Amarouche and Akpinar, 2023, 2024). These constraints have motivated complementary data-driven approaches that learn forecast mappings directly from buoy observations, especially when flexible lead times and stable short-term updates are required for marine operations (De Girolamo et al., 2017; Wang et al., 2023).

2.2. Data-driven wave and metocean forecasting in ocean engineering

Data-driven forecasting in *Ocean Engineering* has been studied extensively for significant wave height (SWH), with recurrent and hybrid models using historical wave and wind observations to produce short-term forecasts (Fan et al., 2020; Gao et al., 2021; Pushpam and Enigo, 2020; Gracia et al., 2021; Minuzzi and Farina, 2023; Zhang et al., 2024). Later studies addressed nearshore variability, data scarcity, and nonstationarity by adding physical or geographic context, transferring knowledge across sites, or decomposing irregular wave signals before recurrent prediction (Jörges et al., 2021; Obara and Nakamura, 2022; Kumar et al., 2018; Dragomiretskiy and Zosso, 2014; Rehman and Aftab, 2019; Zhao et al., 2023; Bekiryazıcı et al., 2025; S. Zhou et al., 2021; Ding et al., 2024; Zheng et al., 2023). Recent marine forecasting work further emphasizes probabilistic uncertainty, interpretability, and robustness under nonstationary conditions, including KDE-based interval forecasting, explainable stacking, and analogous offshore-wind prediction frameworks (Wang and Ying, 2023; Wang et al., 2026, 2024). At the same time, recent SWH-focused critiques caution that complex machine-learning models may not consistently outperform simple autoregressive references, and that time-history plots can reveal delayed tracking even when aggregate errors appear favorable (Jiang et al., 2024; Huang et al., 2023). These studies establish the practical potential of deep learning for operational marine forecasting (De Girolamo et al., 2017; Zis et al., 2020), but they also highlight a gap that is central to the present work: many evaluations remain centered on SWH, fixed lead times, or aggregate errors, whereas operational metocean forecasting requires joint multi-variable prediction, explicit horizon-dependent analysis, and diagnostics that test whether forecasts add useful lead-time information.

2.3. Transformer-based long-horizon forecasting for multivariate metocean series

Transformer-based long-horizon time-series forecasting has improved multi-step sequence modeling through attention, efficient attention variants, and decomposition- or frequency-aware inductive biases (Vaswani et al., 2017; H. Zhou et al., 2021; Wu et al., 2021; Zhou et al., 2022). Patch-based formulations such as PatchTST further reduce the effective sequence length and provide an efficient long-horizon backbone, but applying a generic patching design to buoy-based metocean signals leaves three practical issues: multi-scale periodicity, selective cross-variable interaction, and flexible lead-time conditioning (Nie et al., 2023). Prior LTSF and multivariate forecasting studies motivate partial solutions to these issues, including multi-resolution temporal representations (Wu et al., 2023; Challu et al., 2023), controlled cross-dimension interaction (Zhang et al., 2023), lightweight feature recalibration (Hu et al., 2018), and multi-horizon decoding for heterogeneous temporal inputs (Lim et al., 2021; H. Zhou et al., 2021). In coastal metocean forecasting, these ideas are particularly relevant because waves, winds, currents, and directional variables are coupled but noisy, and because forecasts are often needed across multiple operational lead times. The present study therefore evaluates a compact extension of PatchTST that combines multi-scale patching, gated channel interaction, and horizon-adaptive decoding within a unified multivariate coastal-buoy forecasting framework.

Table 1
Input/output channel schema used in experiments.

Group	Variables (examples)	Encoding	Channels
Numeric metocean	Current speed (m/s); water temp. (°C); Wave height (m); Wind speed (m/s); Air temp. (°C); pressure (hPa)	Real-valued	$C_{\text{num}} = 6$
Directional metocean	Current dir. (deg); Wind dir. (deg)	(sin, cos) for each direction	$C_{\text{dir}} = 4$
Time features	Year/month/day/hour; Day-of-week/day-of-year; Semidiurnal/diurnal/seasonal cycles	Cyclical (sin, cos)	$C_{\text{time}} = 18$
Location features	Latitude, longitude	Real-valued	$C_{\text{loc}} = 2$
Total input			$C_{\text{in}} = 30$
Total output	Numeric + directional targets	Real-valued + (sin, cos)	$C_{\text{out}} = 10$

3. Methodology

3.1. Problem formulation

We consider multivariate time-series forecasting where an input window of length L is mapped to a prediction horizon of length H . Our data are sampled every 30 min; in the main setting we use $L = 720$ and $H = 6$ (15 days of context to predict the next 3 h). The 15-day input window was chosen to cover repeated semidiurnal/diurnal cycles and short synoptic-scale variability while keeping the patch-token length computationally manageable; the 3-hour horizon targets short-term operational forecasting and is kept fixed across models. This $H = 6$ setting defines the main supervised task used for training, while longer recursive-rollout evaluation is introduced later in the experiments. Let $X \in \mathbb{R}^{B \times L \times C_{\text{in}}}$ denote the input sequence and $Y \in \mathbb{R}^{B \times H \times C_{\text{out}}}$ the target sequence, where B is the batch size. Given parameters θ , the model f_θ produces $\hat{Y} = f_\theta(X)$.

Metocean variables and directional encoding. In our setting, the metocean state at time t includes eight physical metocean variables: surface current speed and direction, water temperature, significant wave height, wind speed and direction, air temperature, and air pressure. Directional variables (e.g., wind direction and current direction) are represented using sine/cosine embeddings to avoid discontinuities at $0/2\pi$:

$$(\sin \theta, \cos \theta), \quad \theta \in [0, 2\pi), \quad (1)$$

and are converted back to angles only for evaluation using

$$\hat{\theta} = \text{atan2}(\widehat{\sin \theta}, \widehat{\cos \theta}). \quad (2)$$

Among the eight physical variables, **two are directional (current direction and wind direction) and are encoded as sine/cosine pairs**; therefore, the output target uses six numeric channels plus four directional channels ($C_{\text{out}} = 10$). The full input additionally includes two static station-level location features (latitude and longitude), broadcast along the temporal dimension of the input window, and 18 cyclical calendar/time encoding features, yielding $C_{\text{in}} = 30$ in the standard input configuration (see Table 1).

3.2. Overview of MetoFormer-HQ

MetoFormer-HQ follows the original PatchTST forecasting backbone (Nie et al., 2023) and introduces three modules targeting metocean-specific challenges: (i) multi-scale patch embedding to capture concurrent periodicities, (ii) gated channel interaction to regularize cross-variable information flow under noisy heterogeneous inputs, and (iii) horizon-adaptive query decoding with randomized-horizon training to support flexible lead times and stabilize long-horizon error growth. Fig. 1 summarizes the full pipeline. Fig. 2 provides module-level details (MSPE/GCI/HAQD).

Model configurations used throughout the paper. To keep ablation and result descriptions unambiguous, we use the following names: **BASE** (base configuration): the original PatchTST backbone (Nie et al., 2023); **MS** (Multi-scale extension): BASE + multi-scale patch embedding (MSPE); **MSG** (gated-interaction extension): MS + gated channel interaction (GCI); **HQ** (Horizon-adaptive query extension): MSG + horizon-adaptive query decoding (HAQD) with randomized-horizon training. MetoFormer-HQ and HQ refer to the same final configuration; Section 5 uses the shorthand HQ for compact table/figure reporting. Auxiliary variants HQ-R, HQ-L, and HQ-LR are used only for hard-feature analysis and do not redefine the primary MetoFormer-HQ/HQ configuration.

3.3. Reversible instance normalization (RevIN)

Metocean time series exhibit strong distribution shifts across seasons and weather regimes. We apply reversible instance normalization to each sample on the full input tensor before subsequent modules, including gated channel interaction and patch tokenization (Kim et al., 2022). The normalization covers all $C_{\text{in}} = 30$ input channels (6 numeric metocean, 2 static location, 18 cyclical calendar/time encoding, and 4 directional sine/cosine channels). For an input instance $X \in \mathbb{R}^{L \times C_{\text{in}}}$, we compute sample-wise, channel-wise statistics over the time axis,

$$\mu_c = \frac{1}{L} \sum_{t=1}^L X_{t,c}, \quad \sigma_c = \sqrt{\frac{1}{L} \sum_{t=1}^L (X_{t,c} - \mu_c)^2 + \epsilon}, \quad (3)$$

and normalize as

$$X'_{t,c} = \frac{X_{t,c} - \mu_c}{\sigma_c}. \quad (4)$$

After predicting $\hat{Y}'$ in the normalized space, we restore the original scale only for the predicted target subset ($C_{\text{out}} = 10$: 6 numeric + 4 directional channels) by $\hat{Y}_{t,c} = \hat{Y}'_{t,c} \sigma_c + \mu_c$, where (μ_c, σ_c) are taken from the corresponding input-side target channels. This is used to mitigate distribution-shift effects without changing the backbone architecture.

3.4. Multi-scale patch embedding (MSPE)

Using the notation defined in Section 3, patch-based forecasting represents the input as a sequence of tokens extracted from local temporal windows. To capture concurrent periodicities in metocean observations, MSPE constructs patch tokens at multiple temporal scales. For each scale $s \in \{1, \dots, S\}$ with patch length P_s and stride S_s , we extract N_s patches:

$$N_s = \left\lfloor \frac{L - P_s}{S_s} \right\rfloor + 1, \quad \mathbf{p}_i^{(s)} \in \mathbb{R}^{P_s \times C_{\text{in}}}, \quad i = 1, \dots, N_s. \quad (5)$$

Each patch is mapped to a d -dimensional token via a scale-specific learnable 1D convolutional projection $\phi_s(\cdot)$:

$$\mathbf{z}_i^{(s)} = \phi_s(\mathbf{p}_i^{(s)}) \in \mathbb{R}^d. \quad (6)$$

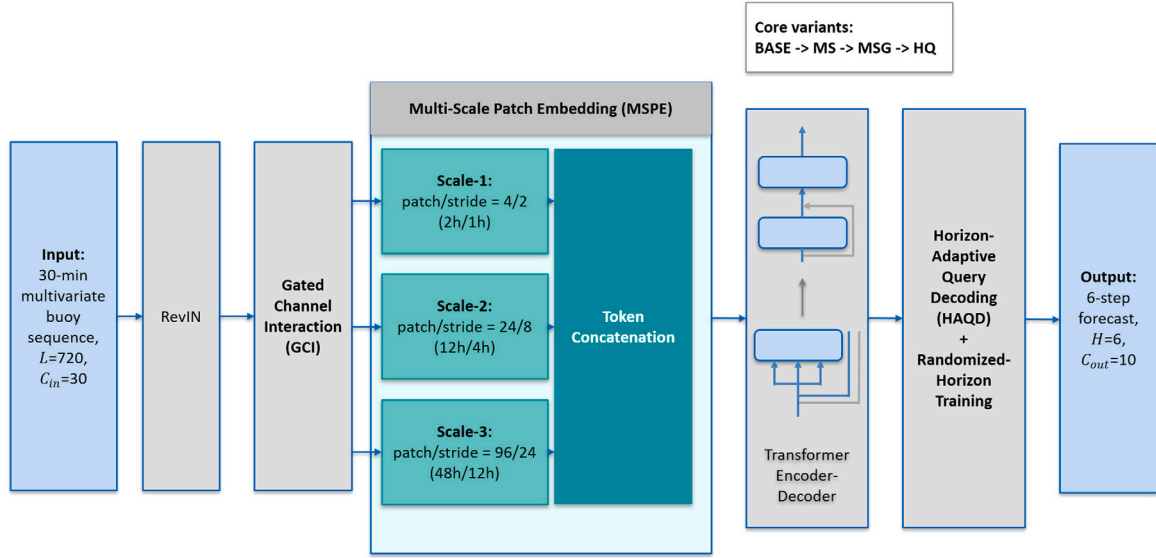


Fig. 1. Overview of MetoFormer-HQ with MSPE, GCI, HAQD, and randomized-horizon training.

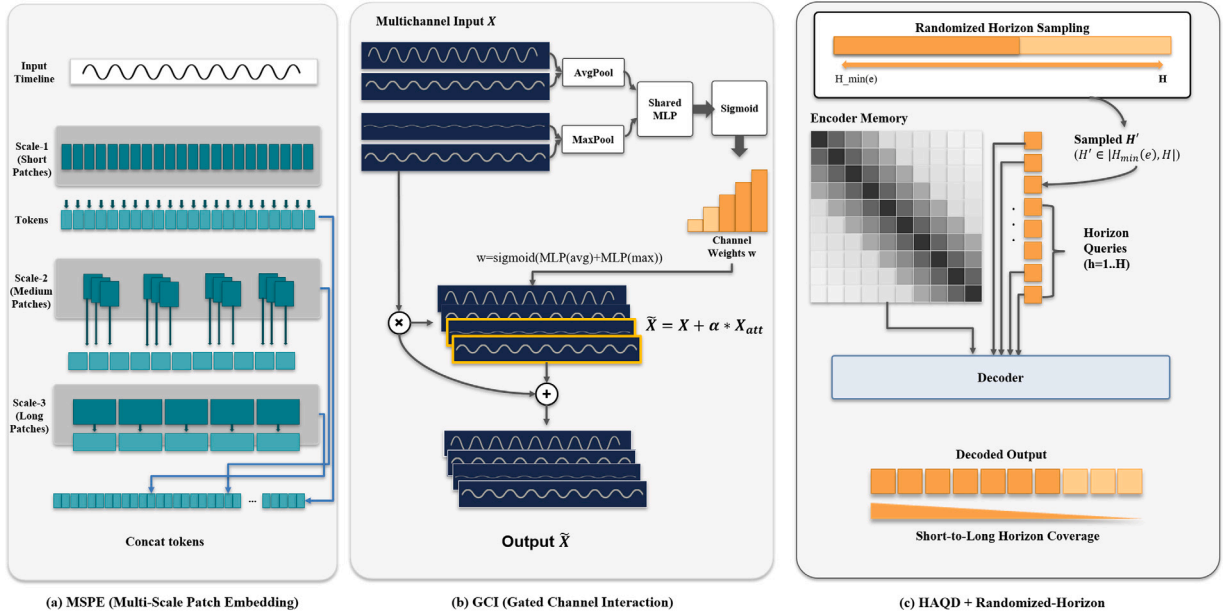


Fig. 2. Module mechanisms for (a) MSPE, (b) GCI, and (c) HAQD.

In MS, we fuse multi-scale tokens into a unified sequence using token concatenation with scale embeddings:

$$\bar{\mathbf{z}}_i^{(s)} = \mathbf{z}_i^{(s)} + \mathbf{e}_s, \quad \mathbf{Z} = \text{Concat}(\{\bar{\mathbf{z}}_i^{(s)}\}_{i=1}^{N_s})_{s=1}^S. \quad (7)$$

In the main MS-HQ configurations on 30-minute data, we use three scales $(P_s, S_s) \in \{(4, 2), (24, 8), (96, 24)\}$, which correspond to receptive-field/stride pairs of 2 h/1 h, 12 h/4 h, and 48 h/12 h. This three-scale setting is a practical minimal hierarchy to cover short-, intermediate-, and long-range meteocean variability, based on domain knowledge and preliminary tuning. In our setup, two scales were insufficient to clearly separate sub-daily/tidal patterns from multi-day synoptic variability, whereas four or more scales increased token count and redundant information with uncertain benefit relative to added computation. The resulting token sequence $\mathbf{Z}$ is consumed by the Transformer encoder.

3.5. Gated channel interaction (GCI)

Metocean variables are physically coupled, but their cross-variable dependencies can be intermittent and corrupted by noise. GCI is applied before patch embedding as a lightweight gated channel reweighting module for conservative cross-channel interaction, using pooled channel descriptors. Given $\mathbf{X} \in \mathbb{R}^{L \times C_{in}}$, we compute average and max pooled descriptors over time:

$$\mathbf{s}_{avg} = \text{AvgPool}_t(\mathbf{X}), \quad \mathbf{s}_{max} = \text{MaxPool}_t(\mathbf{X}), \quad (8)$$

then obtain channel weights with a shared MLP:

$$\mathbf{w} = \sigma(\text{MLP}(\mathbf{s}_{avg}) + \text{MLP}(\mathbf{s}_{max})) \in \mathbb{R}^{C_{in}}. \quad (9)$$

The shared MLP is a shared two-layer bottleneck module with reduction ratio 4, i.e., $\text{Linear}(n_{channels}, n_{reduced})\text{-ReLU-Linear}(n_{reduced}, n_{channels})$.

with $n_{\text{reduced}} = n_{\text{channels}}/4$, and the same MLP is applied to both pooled descriptors. Channel-attended inputs are

$$\mathbf{X}_{\text{att}} = \mathbf{X} \odot \mathbf{w}. \quad (10)$$

A single global learnable scalar gate (shared across channels) controls add-mode residual mixing in all reported GCI-enabled experiments:

$$\alpha = \sigma(g/\tau), \quad \tilde{\mathbf{X}} = \mathbf{X} + \alpha \mathbf{X}_{\text{att}}. \quad (11)$$

where g is a learnable parameter and τ is a temperature. In this study, GCI is designed primarily as a stabilizing reweighting mechanism. We log and aggregate gate activations (e.g., mean values over batches and selected regimes) for interpretability analyses in Section 5 (Fig. 7).

3.6. Horizon-adaptive query decoding (HAQD) and randomized-horizon training

Operational forecasting requires flexible lead times, and fixed-horizon training may over-specialize a model to a single prediction length. HAQD addresses this by conditioning decoding on horizon steps via dynamic time queries, with one query assigned to each forecast step. For each horizon step h , we use a sinusoidal query embedding:

$$\mathbf{q}_h = \text{PE}(h) \in \mathbb{R}^d, \quad (12)$$

where $\text{PE}(\cdot)$ denotes a standard sinusoidal positional encoding for query conditioning. Collecting all steps gives the decoder query sequence $\{\mathbf{q}_h\}_{h=1}^H$. Let $\mathbf{M} \in \mathbb{R}^{T \times d}$ denote the shared encoder memory. These queries act as decoder targets and attend to $\mathbf{M}$ through query-conditioned cross-attention, producing step-specific decoder states that are mapped to output channels by a single shared output projection head applied identically across all horizon steps.

Randomized-horizon training. To improve performance consistency across lead times, we train with randomized horizons. During training, the lower bound of sampled horizons is scheduled by warmup and ramp epochs, and loss is computed on the first H' steps.

Algorithm 1. Randomized-horizon training (HAQD).

1. For epoch e , compute the minimum sampled horizon $H_{\min}(e)$: warmup uses $H_{\min}(e) = H$, then linearly ramps from $H_{\min}^{\text{start}}$ to $H_{\min}^{\text{final}}$.
2. Sample $H' \sim \mathcal{U}\{H_{\min}(e), \dots, H\}$ and set horizon steps $\{1, \dots, H'\}$.
3. Decode only these H' steps and compute loss on $(\hat{Y}_{:,1:H'}, Y_{:,1:H'})$.
4. Update θ (optionally with horizon-dependent loss weights).

3.7. Training objective

The default objective is MSE over all active horizon steps and all output channels:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{BH'C_{\text{out}}} \sum_{b=1}^B \sum_{h=1}^{H'} \sum_{c=1}^{C_{\text{out}}} (Y_{b,h,c} - \hat{Y}_{b,h,c})^2. \quad (13)$$

When hard-feature robust training is enabled (used only in hard-feature extension runs), hard channels $I_{\text{hard}} = \{0, 2, 3\}$ (current speed, wave height, wind speed) are optimized with SmoothL1 (Huber) while other channels keep MSE:

$$\ell_{b,h,c} = \begin{cases} \text{SmoothL1}(Y_{b,h,c}, \hat{Y}_{b,h,c}; \delta), & c \in I_{\text{hard}} \\ (Y_{b,h,c} - \hat{Y}_{b,h,c})^2, & \text{otherwise} \end{cases} \quad (14)$$

$$\mathcal{L}_{\text{robust}} = \frac{1}{BH'C_{\text{out}}} \sum_{b,h,c} \omega_c \ell_{b,h,c}, \quad \omega_c = \begin{cases} \beta, & c \in I_{\text{hard}} \\ 1, & \text{otherwise} \end{cases} \quad (15)$$

where δ is the Huber threshold and β is a hard-channel boost factor.

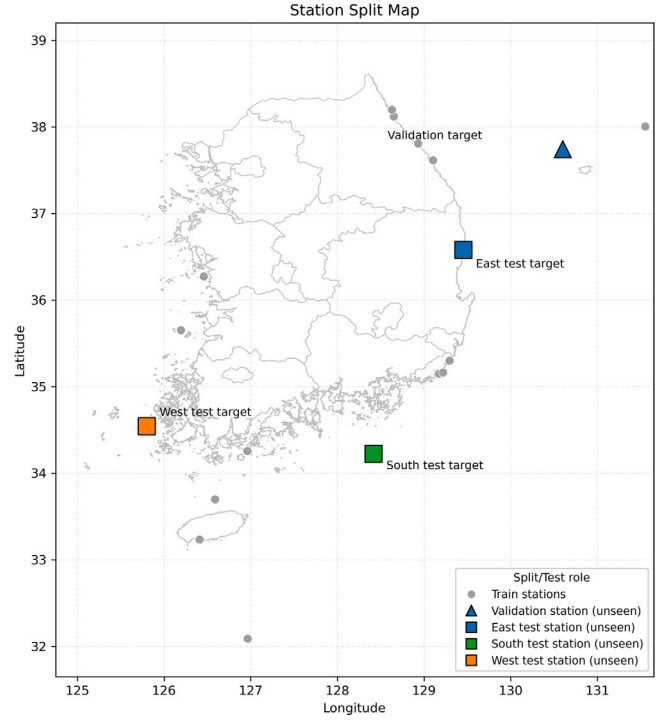


Fig. 3. Spatial OOD station split across East, South, and West coastal regions.

4. Experiments

4.1. Dataset and preprocessing

We use time series collected by coastal buoy stations around Korea. Raw buoy observations were obtained from the Korea Hydrographic and Oceanographic Agency (KHOA) OceanGrid observation portal (<http://www.khoa.go.kr/oceangrid/gis/category/observe/observeSearch.do>, accessed March 10, 2026). The dataset includes current speed/direction, water temperature, significant wave height, wind speed/direction, air temperature, air pressure, and station coordinates. Directional variables are encoded using sine/cosine components, while the headline error metrics are computed on the six scalar variables after inverse normalization. The sampling interval is 30 min. After download, the raw records were reprocessed by the authors into 30-minute files and the train/validation/test splits used in this study. Validation and test stations are unseen during training (spatial OOD evaluation). The summarized split description below corresponds to one unseen validation station and one unseen test station. Table 2 reports the split-level audit (rows, stations, period, and missingness before/after preprocessing), and Fig. 3 visualizes the station-level spatial split. The train split contains 1,910,112 rows from 16 stations (2018-01-01 to 2025-04-15), the validation split contains 127,729 rows from one unseen station, and the test split contains 50,614 rows from another unseen station (2022-05-26 to 2025-04-15). Missing values are processed by time-based interpolation, then forward/backward filling. Residual numeric missing values are replaced by column means; this pipeline removes residual NaNs in all three splits. Normalization statistics are estimated from the training split and then applied to validation and test splits to avoid leakage. The same single-station spatial OOD protocol is applied consistently to three regional unseen test targets (East/South/West), with one unseen test station per region under the same training/evaluation procedure. Regional reporting uses a unified East/South/West format under identical protocol settings. Unless noted otherwise, experiments use input length $L = 720$ and horizon $H = 6$.

Table 2

Dataset split summary and missingness audit.

Split	Stns.	Rows	Period (start/end)	Raw miss. (%)	Post-imp. miss. (%)
Train	16	1.91M	2018-01-01 2025-04-15	15.38	0.00
Validation	1	127.7k	2018-01-01 2025-04-15	28.55	0.00
Test	1	50.6k	2022-05-26 2025-04-15	17.40	0.00
Total	18	2.09M	2018-01-01 2025-04-15	16.23	0.00

4.2. Baselines and implementation details

We compare against representative linear and deep time-series forecasting baselines, including DLinear/NLinear and the hierarchical baseline N-HiTS (Zeng et al., 2023; Challu et al., 2023). To address comparison with a recent Transformer-family forecaster, we additionally evaluate iTransformer, which treats variates as tokens through an inverted Transformer formulation (Liu et al., 2024). Because iTransformer is an external comparator rather than a component of the proposed ablation chain, its results are reported separately from the core BASE-HQ table. We quantify incremental gains by progressively extending the original PatchTST configuration, denoted BASE, to HQ (Nie et al., 2023). To diagnose short-horizon autocorrelation and delayed-tracking behavior, we also include two intentionally simple references: a persistence baseline and a feature-wise AR(1) baseline. The persistence forecast repeats the last observed value for every future lead,

$$\hat{x}_{t+h,c}^{\text{per}} = x_{t,c}, \quad h = 1, \dots, H, \quad (16)$$

where c denotes the scalar variable. The AR(1) baseline fits a separate one-step linear autoregression for each scalar variable on the training split and then iterates it over the forecast horizon. These baselines are not intended as architectural competitors; they are included to diagnose short-horizon autocorrelation and delayed-tracking behavior.

Experimental configurations. To ensure a consistent ablation narrative, we define four incremental configurations: **BASE**: the original PatchTST configuration (Nie et al., 2023); **MS**: BASE + multi-scale patch embedding; **MSG**: MS + gated channel interaction; **HQ**: MSG + dynamic time queries with randomized-horizon training. For hard-feature ablation runs, we additionally use **HQ-R** (HQ + robust hard-feature objective), **HQ-L** (HQ + lag-aware input augmentation), and **HQ-LR** (HQ + lag-aware augmentation + robust hard-feature objective). These are auxiliary variants for hard-feature analysis; they do not redefine the primary model, MetoFormer-HQ.

All models are trained with Adam and a OneCycle learning-rate schedule (Kingma and Ba, 2015; Smith, 2018). Maximum epoch and batch-size settings were selected by model family using validation-only runs for stable convergence under the available GPU memory. DLinear, NLinear, and N-HiTS baselines are trained for up to 20 epochs with batch size 32, whereas BASE-HQ and hard-feature extension runs are trained for up to 50 epochs with batch size 128. The iTransformer external comparator uses the same maximum epoch budget and batch size as the main learned models. These are maximum training limits; in all settings, early stopping is based on validation RMSE with patience 5. For MS-HQ we use $d_{\text{model}} = 128$, 8 attention heads, 3 encoder layers, 2 decoder layers, feedforward dimension 512, and dropout 0.1. Unless otherwise noted, we report mean and standard deviation over three seeds (42, 43, 44). For GCI-enabled runs (MSG-HQ), settings are gate initialization 1.0, temperature 1.0, clip 2.0, dropout 0.05, warmup 500 steps, and additive residual mode. For HAQD randomized-horizon training (HQ and variants), settings are $H_{\min}^{\text{start}} = 4$, $H_{\min}^{\text{final}} = 1$, warmup 5 epochs, ramp 10 epochs, linear horizon weighting (decay 0.5), and query-norm clip 8.0. Hyperparameter selection was performed using

Table 3

Reproducibility checklist for experiments.

Recorded items
• Hardware/software: RTX A6000, Threadripper 3960X, 128 GB RAM; Python 3.9, PyTorch 2.0.1+cu118, NumPy 1.23.5, pandas 2.2.3, scikit-learn 1.6.1. • Data/split: 16 training stations, one validation station, and one East OOD test station; 30-min sampling; normalization fit only on the training split. • Forecast protocol: direct task uses $L = 720$ and $H = 6$; longer-lead checks use recursive rollout buckets; metrics are computed after inverse normalization. • Reporting: stochastic learned models use seeds 42–44; checkpoints are selected by validation RMSE with patience 5; results are reported as mean $\pm$ standard deviation where applicable. • Cost scale: mean training times range from 43 min (linear baselines) to 582 min (HQ); trainable parameters range from 4.6k to 2.34M across reported models.

Table 4

Hyperparameter-selection checklist.

Validation-only selection records
• Common protocol: fixed OOD split, $L = 720$, $H = 6$, scaler, metrics, and seed policy; no test-station tuning. • MetoFormer-HQ family: Adam/OneCycle, max 50 epochs, batch 128, learning rate 10^{-4}, patience 5, $d_{\text{model}} = 128$, 8 heads, 3 encoder layers, 2 decoder layers, FFN 512, dropout 0.1. • Module settings: MSPE, GCI, HAQD, hard-feature loss, and randomized-horizon settings were selected from validation-only stability/performance checks and frozen before final testing. • Learned baselines: DLinear, NLinear, and N-HiTS used the same split/scaler/metric protocol with validation-RMSE early stopping, max 20 epochs, batch 32, and learning rate 10^{-4}. • iTransformer: budget-matched external comparator with max 50 epochs, batch 128, learning rate 10^{-4}, patience 5, and the same hidden-size/head/layer scale as the main Transformer runs. • Naive diagnostics: persistence and feature-wise AR(1) are diagnostic references fit without test-set tuning.

the validation station only; the unseen test station was not used for architecture selection, early stopping, or post-hoc tuning. Shared training parameters (optimizer, learning-rate scale, input length, horizon, scaler, split files, seeds, and metric definitions) were kept fixed across the main learned models whenever applicable. For external baselines, we used recommended or commonly adopted configurations as the starting point and then selected among small validation-only variants under the same early-stopping rule, rather than tuning separately on the test results. Architecture-specific parameters introduced by MS, MSG, and HQ were selected from preliminary validation runs to balance validation RMSE, training stability, and computational feasibility; they were then frozen before final East/South/West test evaluation. Table 3 summarizes the computational environment, split files, seed policy, hyperparameter-selection policy, recorded training times, and parameter counts used for the reproducibility audit. Table 4 further audits the main model and baseline hyperparameters, distinguishing fixed/common settings from validation-only selections and explicitly recording that the unseen test station was not used for tuning.

4.3. Evaluation protocols and metrics

We report standard forecasting metrics: RMSE, MAE, and R^2 ,

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}, \quad (17)$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|, \quad (18)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}. \quad (19)$$

Here, y_i , $\hat{y}_i$, and $\bar{y}$ denote the observed value, predicted value, and sample mean of the observed values over N evaluated samples, respectively. In addition to overall scores, we report horizon-wise metrics

($t+1$ to $t+6$), feature-wise metrics on the six numeric channels, and peak-event precision/recall/F1 using a 90th-percentile threshold. To examine longer-lead behavior, we also report recursive-rollout metrics for both the baselines and the MetoFormer family using lead buckets ($t+1\sim6$, $t+7\sim12$, $t+13\sim24$, $t+25\sim36$). The direct multi-step setting evaluates the native $H = 6$ output, whereas the recursive setting repeatedly rolls the model forward beyond the direct horizon. Directional channels are predicted during training and inference, while the headline tables focus on the six scalar variables most relevant for operational interpretation. Regional metrics are reported under identical metric definitions and seed settings. To quantify uncertainty in the main model comparisons, we compute paired moving-block bootstrap confidence intervals over ordered test anchors. The block length is 48 anchors (24 h at 30-minute sampling), with 1000 bootstrap replicates; stochastic model runs are resampled by seed position within each bootstrap replicate.

For timing-focused evaluation, we further compute cross-correlation-based lag, first-difference correlation, first-difference direction accuracy, and peak-window matching for wave height. For a candidate lag $\ell \in [-K, K]$, with $K = 6$ in this study, we compute the lagged correlation

$$\rho(\ell) = \text{corr}(y_{1:T-\ell}, \hat{y}_{1+\ell:T}), \quad (20)$$

for $\ell \geq 0$; the analogous overlapping segments are used for $\ell < 0$. We report the lag that maximizes $\rho(\ell)$ within the six-step search window. Under this definition, a positive best lag indicates delayed tracking because the prediction sequence must be shifted forward to align with observations. Change-based metrics are computed from $\Delta y_t = y_{t+1} - y_t$ and $\Delta \hat{y}_t = \hat{y}_{t+1} - \hat{y}_t$ along the ordered test anchors:

$$\rho_A = \text{corr}(\Delta y_{1:T-1}, \Delta \hat{y}_{1:T-1}), \quad (21)$$

$$\text{DirAcc} = \frac{1}{T-1} \sum_{t=1}^{T-1} \mathbf{1}\{\text{sign}(\Delta y_t) = \text{sign}(\Delta \hat{y}_t)\}. \quad (22)$$

For peak-window matching, local peaks above the 90th percentile are extracted separately from observed and predicted wave-height sequences. If $\mathcal{P}$ is the set of observed peaks and $\hat{\mathcal{P}}$ is the set of predicted peaks, the match rate within a radius $r = 6$ steps is

$$\text{PeakMatch} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \mathbf{1}\{\exists \hat{p} \in \hat{\mathcal{P}} : |\hat{p} - p| \leq r\}. \quad (23)$$

For event-focused thresholding, precision, recall, and F1 are computed as

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (24)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (25)$$

$$\text{F1} = \frac{2 \text{Precision} \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (26)$$

where positives are defined by the 90th-percentile event threshold.

5. Results and discussion

5.1. Overall multi-step forecasting performance on Korean buoy observations

Unless otherwise stated, metrics in this subsection are reported on the East single-station OOD split. For readability, abbreviation suffixes indicate added modules: MS (multi-scale), G (gated interaction), HQ (horizon-adaptive query), R (robust hard-feature objective), and L (lag-aware augmentation). In this section, HQ denotes the final MetoFormer-HQ configuration. Table 5 summarizes overall performance. Among trained neural/linear baselines, N-HiTS has the lowest RMSE and MAE (4.5830 and 2.5419). BASE (original PatchTST) improves over these trained baselines (RMSE 3.4025, MAE 1.7747), and HQ yields the lowest RMSE among the core configurations (3.2071). Relative to BASE,

Table 5

East single-station OOD overall performance (mean $\pm$ std, $n = 3$ seeds).

Model	n	RMSE $\downarrow$	MAE $\downarrow$	R^2 $\uparrow$
N-HiTS	3	4.5830 $\pm$ 1.0453	2.5419 $\pm$ 0.7039	0.6417 $\pm$ 0.1206
NLinear	3	5.9165 $\pm$ 0.1030	3.0719 $\pm$ 0.0479	0.4961 $\pm$ 0.0088
DLinear	3	5.8803 $\pm$ 0.7270	3.0931 $\pm$ 0.3690	0.4849 $\pm$ 0.1003
BASE	3	3.4025 $\pm$ 0.0043	1.7747 $\pm$ 0.0137	0.7139 $\pm$ 0.0148
MS	3	3.4278 $\pm$ 0.1065	1.7010 $\pm$ 0.0884	0.7315 $\pm$ 0.0437
MSG	3	3.4016 $\pm$ 0.1278	1.7219 $\pm$ 0.0935	0.7246 $\pm$ 0.0316
HQ	3	3.2071 $\pm$ 0.0414	1.5952 $\pm$ 0.1001	0.7542 $\pm$ 0.0330
HQ-R	3	3.2262 $\pm$ 0.1051	1.5403 $\pm$ 0.0835	0.7682 $\pm$ 0.0245
HQ-L	3	3.4114 $\pm$ 0.0776	1.7025 $\pm$ 0.1012	0.7397 $\pm$ 0.0320
HQ-LR	3	3.4103 $\pm$ 0.0610	1.6480 $\pm$ 0.0832	0.7514 $\pm$ 0.0217

HQ reduces RMSE by 5.74% and MAE by 10.11%; relative to N-HiTS, HQ reduces RMSE by 30.02% and MAE by 37.24%. Among auxiliary hard-feature variants, HQ-R yields the lowest MAE (1.5403) and highest R^2 (0.7682), with RMSE close to HQ (3.2262 vs. 3.2071). Taken together, these results indicate that the original PatchTST backbone is already a strong temporal prior for metocean forecasting, and that the gains of MetoFormer-HQ come from adapting that backbone to multi-scale periodicity and flexible-horizon deployment rather than replacing it with a different forecasting family. The auxiliary variants further suggest a practical trade-off between minimizing aggregate error and emphasizing difficult high-variance channels, which is important when operational priorities depend on specific variables rather than only on global averages.

Table 6 adds persistence and AR(1) references for the East single-station OOD direct forecast. The persistence baseline is deliberately stringent in this short-horizon setting because 30-minute buoy observations, especially wave height, are highly autocorrelated. Its strong aggregate RMSE/MAE therefore does not invalidate the proposed architecture, but it does show that aggregate error alone is insufficient to prove useful lead-time prediction. For this reason, the following sections interpret the proposed model using horizon-wise, lag-sensitive, peak-event, and representative time-history diagnostics rather than only a single global error score.

5.2. Regional single-station OOD summary

Table 7 reports regional single-station OOD results for core models (BASE-HQ) under the unified protocol and seed policy (mean $\pm$ std over three seeds). In East, HQ is best across all three metrics (RMSE 3.2071, MAE 1.5952, R^2 0.7542). In South, HQ records the lowest RMSE/MAE and highest R^2 (7.3978/2.7375/0.8324), with a small RMSE margin over MSG (7.4015). In West, HQ remains competitive in RMSE/MAE and attains the highest explanatory power (R^2 0.6744) within the core ablation set. These regional outcomes show broadly consistent learned-model behavior across East, South, and West targets under the same training/evaluation protocol. This pattern suggests that the proposed modules are not exploiting only one site-specific regime, but remain informative across heterogeneous coastal environments. The larger West-target errors may be partly related to the shallow, tidally energetic, and bathymetrically complex nature of the Korean western coast. Such local wave-current and water-level effects are not explicitly represented in the present station-wise temporal inputs, which supports the need for future tide, bathymetry, atmospheric forcing, and spatial-context variables. At the same time, the remaining regional gap in absolute error indicates that unresolved local forcing and coastal dynamics still matter, motivating future spatial or exogenous conditioning beyond the current station-wise setup. Table 8 reports an additional budget-matched comparison with iTransformer under the same East/South/West single-station OOD protocol. This comparison is intentionally separated from Table 7: Table 7 tests the incremental contribution of the proposed modules, whereas Table 8 addresses whether the final model remains competitive against a recent

Table 6
Naive diagnostic baselines for East OOD direct forecasting.

Model	Role	n	RMSE ↓	MAE ↓	R^2 ↑
Persistence	Last-observation repeat	1	2.8432	1.0184	0.8586
HQ	Proposed model	3	3.2070 ± 0.0415	1.5952 ± 0.1002	0.7541 ± 0.0330
BASE	Original PatchTST	3	3.4025 ± 0.0043	1.7747 ± 0.0137	0.7139 ± 0.0148
AR(1)	Feature-wise autoregression	1	4.2217	1.9661	0.7505
N-HiTS	Neural baseline	3	4.5808 ± 1.0437	2.5412 ± 0.7034	0.6417 ± 0.1206

Table 7
Regional single-station OOD performance of core models.

Model	Region	RMSE ↓	MAE ↓	R^2 ↑
BASE	East	3.4025 ± 0.0043	1.7747 ± 0.0137	0.7139 ± 0.0148
	South	7.4638 ± 0.3417	2.9613 ± 0.1877	0.7932 ± 0.0306
	West	15.6608 ± 0.3041	5.7437 ± 0.0801	0.6503 ± 0.0057
MS	East	3.4278 ± 0.1065	1.7010 ± 0.0884	0.7315 ± 0.0437
	South	7.4372 ± 0.0994	2.7721 ± 0.0938	0.8289 ± 0.0163
	West	15.6783 ± 0.4353	5.6383 ± 0.1339	0.6538 ± 0.0324
MSG	East	3.4016 ± 0.1278	1.7219 ± 0.0935	0.7246 ± 0.0316
	South	7.4015 ± 0.0494	2.7748 ± 0.1032	0.8235 ± 0.0136
	West	16.2447 ± 0.5039	5.7437 ± 0.1818	0.6618 ± 0.0181
HQ	East	3.2071 ± 0.0414	1.5952 ± 0.1001	0.7542 ± 0.0330
	South	7.3978 ± 0.0894	2.7375 ± 0.1357	0.8324 ± 0.0203
	West	16.1850 ± 0.6528	5.7275 ± 0.1521	0.6744 ± 0.0253

Table 8
Budget-matched iTransformer comparison on regional OOD splits.

Region	Model	RMSE ↓	MAE ↓	R^2 ↑
East	BASE	3.4025 ± 0.0043	1.7747 ± 0.0137	0.7139 ± 0.0148
	HQ	3.2071 ± 0.0414	1.5952 ± 0.1001	0.7542 ± 0.0330
	iTransformer	3.4111 ± 0.4213	1.6062 ± 0.2284	0.7690 ± 0.0662
South	BASE	7.4638 ± 0.3417	2.9613 ± 0.1877	0.7932 ± 0.0306
	HQ	7.3978 ± 0.0894	2.7375 ± 0.1357	0.8324 ± 0.0203
	iTransformer	7.8079 ± 0.8444	2.8213 ± 0.4245	0.8333 ± 0.0476
West	BASE	15.6608 ± 0.3041	5.7437 ± 0.0801	0.6503 ± 0.0057
	HQ	16.1850 ± 0.6528	5.7275 ± 0.1521	0.6744 ± 0.0253
	iTransformer	14.3839 ± 0.1132	5.1901 ± 0.0515	0.7597 ± 0.0060

Transformer-family forecaster. The result is mixed and region-dependent. HQ retains the lowest RMSE and MAE in East and South, while iTransformer has slightly higher R^2 in those two regions. In West, iTransformer performs better than both HQ and BASE on RMSE, MAE, and R^2 . Thus, the additional comparison supports a region-dependent interpretation: HQ is competitive with a strong recent comparator, but the relative advantage depends on the regional target. This motivates the conservative interpretation used in this manuscript: the main contribution is the module-wise improvement and diagnostic forecasting behavior of MetoFormer-HQ, rather than a broad architectural ranking against every Transformer-family model.

5.3. Long-horizon skill and error growth analysis

This subsection reports East single-station OOD long-horizon behavior.

As shown in Fig. 4 and Table 9, HQ achieves the lowest RMSE at every horizon among BASE–HQ (from 2.4722 at $t+1$ to 3.7392 at $t+6$). At the longest lead time ($t+6$), HQ remains better than BASE (3.8467) and N-HiTS (4.8289). HQ-R (auxiliary hard-feature variant) performs best at short lead times ($t+1$ to $t+2$) and remains competitive at long lead times. Table 10 reports extended-horizon recursive-rollout results using lead buckets ($t+1 \sim 6$, $t+7 \sim 12$, $t+13 \sim 24$, $t+25 \sim 36$). Under this recursive-bucket protocol, the learned baselines show clear long-horizon error accumulation (e.g., N-HiTS RMSE $4.5809 \rightarrow 7.2042$; NLinear and DLinear reach negative R^2 at $t+25 \sim 36$), whereas MetoFormer variants maintain lower errors and positive R^2 in the same horizon ranges. Within the proposed learned-model family, HQ is best

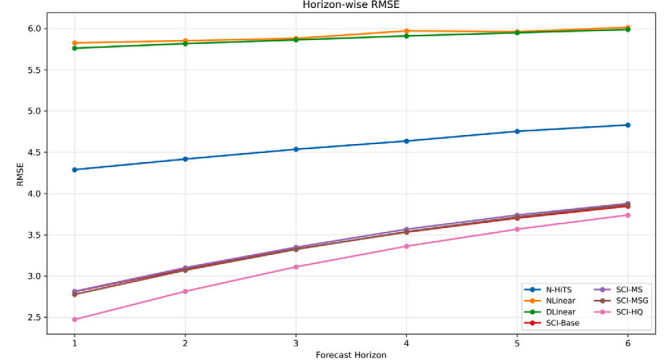


Fig. 4. Horizon-wise RMSE by lead time.

through $t+24$ on all three metrics and remains competitive at $t+25 \sim 36$. This pattern suggests that, relative to BASE and the selected learned baselines, MetoFormer-HQ reduces error growth under this recursive-rollout evaluation. The positive R^2 values in the farthest lead buckets suggest that the forecasts retain useful signal under this protocol rather than collapsing toward a weak mean predictor, although the narrowing margin at $t+25 \sim 36$ also implies that a temporal-only setup eventually reaches a predictability limit. Fig. 5 indicates that HQ achieves the lowest RMSE across horizons on four channels (current speed, wind speed, air temperature, and air pressure). For wave height and water temperature, baseline models remain competitive and can yield lower RMSE at the same horizons. The feature-wise split therefore suggests that horizon adaptation is especially beneficial for rapidly varying atmospheric or kinematic channels, whereas smoother variables can still be served reasonably well by simpler baselines.

5.4. Lag-sensitive diagnostics for delayed-tracking behavior

A key concern in short-horizon metocean forecasting is that a model may simply track delayed observations, especially for wave height, where persistence can be very strong. Table 11 therefore reports wave-height timing diagnostics for BASE, HQ, and persistence at representative direct leads ($t+1$, $t+3$, and $t+6$). The best-lag values confirm that delayed tracking is a real risk: BASE reaches its highest cross-correlation at the maximum six-step lag for all reported leads. HQ does not remove this effect entirely, but it improves zero-lag correlation at every reported lead (for example, 0.846 vs. 0.771 at $t+1$ and 0.773 vs. 0.731 at $t+6$) and increases peak-window matching relative to BASE. Persistence has the strongest aggregate error in Table 6, but its diagnostic interpretation is different: by construction, it repeats the latest observation and therefore represents a conservative no-change reference rather than a learned lead-time mapping (see Fig. 6).

The raw-amplitude time-history view is intentionally retained because wave-height amplitude is operationally important. It also shows a remaining limitation: HQ reduces the strong smoothing seen in BASE but still underestimates the largest peak amplitudes at longer direct leads. We therefore use this figure as a diagnostic example of timing and event-window behavior, not as evidence that peak amplitude errors are fully resolved.

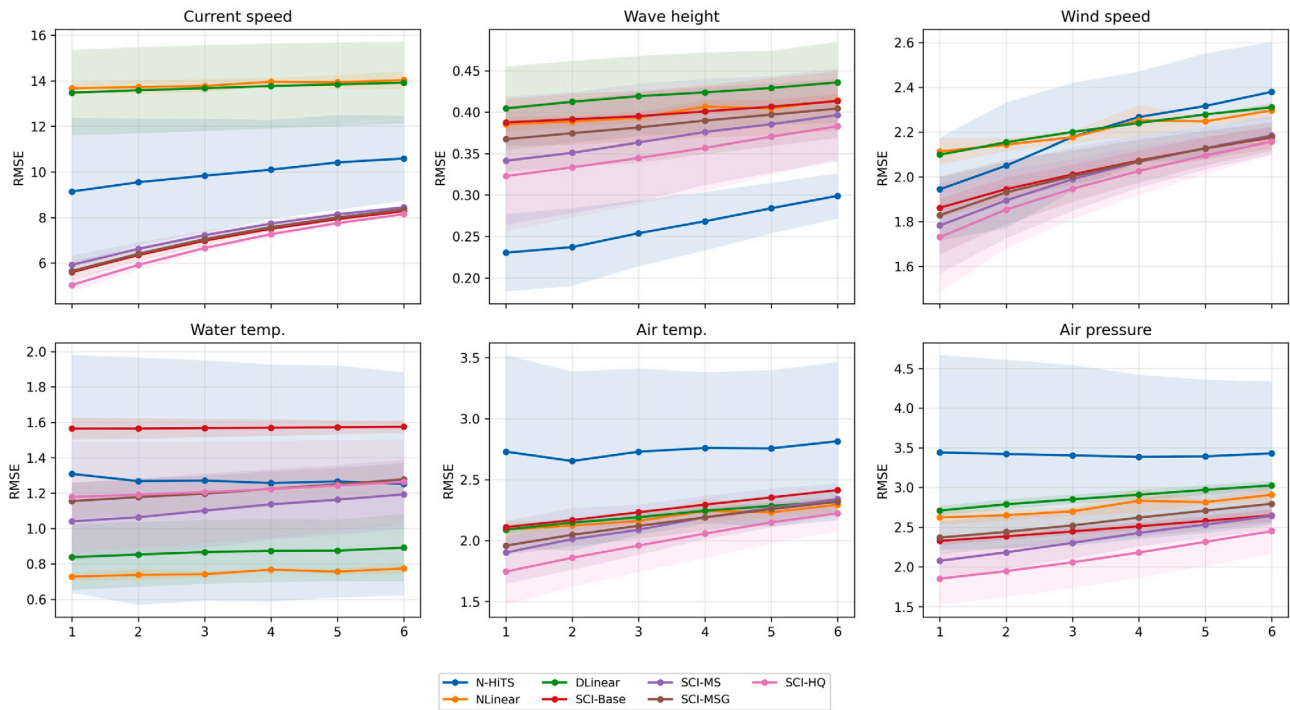
Horizon-wise RMSE by Feature (mean $\pm$ std across seeds)

Fig. 5. Feature-wise horizon RMSE for the core model set.

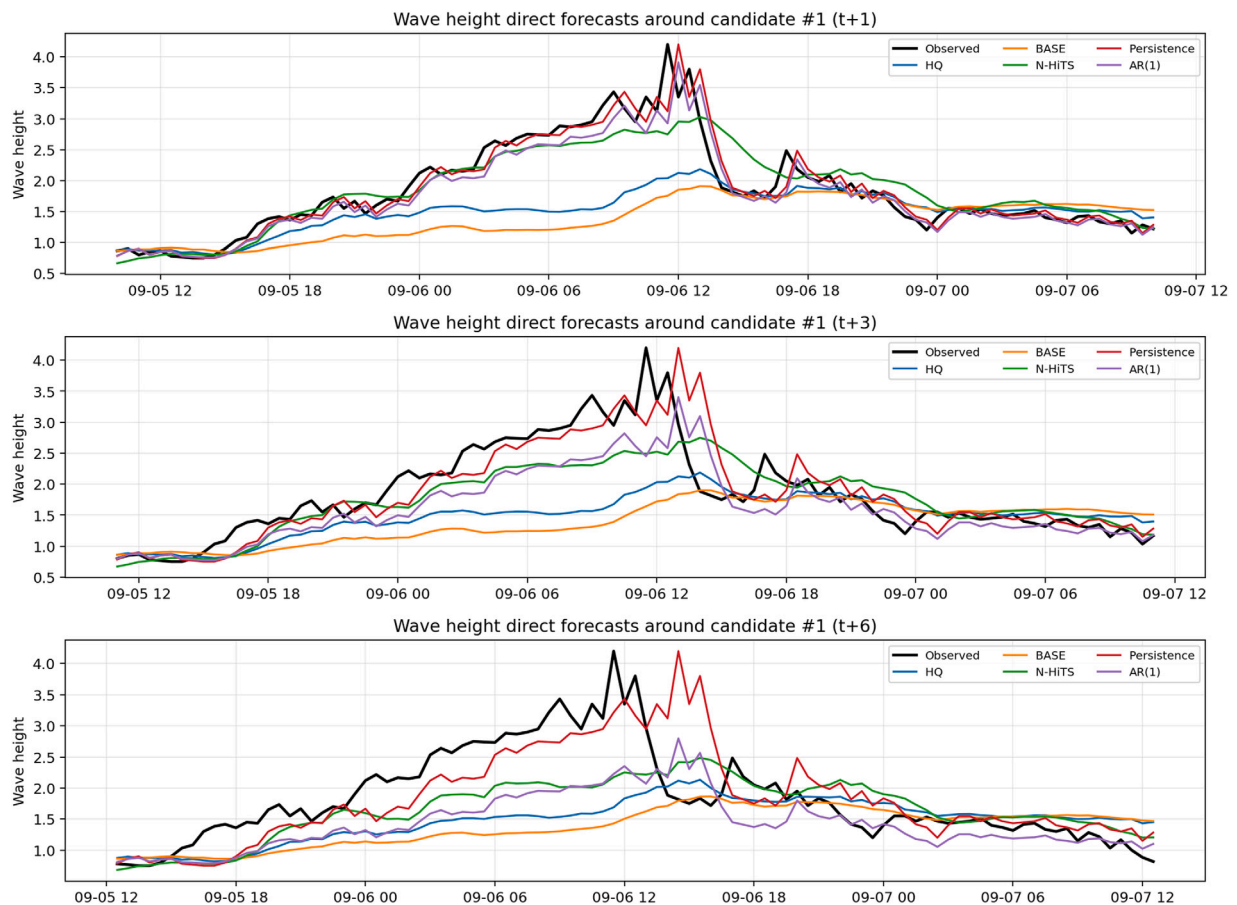


Fig. 6. Representative East OOD wave-height time-history diagnostic for a high-change peak-event window.

Table 9
East OOD horizon-wise RMSE at $t+1$ – $t+6$.

Model	$t+1$	$t+2$	$t+3$	$t+4$	$t+5$	$t+6$
N-HITS	4.2867 ± 1.3296	4.4154 ± 1.2095	4.5338 ± 1.0795	4.6341 ± 0.9536	4.7519 ± 0.9300	4.8289 ± 0.8471
BASE	2.8094 ± 0.0171	3.0857 ± 0.0073	3.3270 ± 0.0037	3.5318 ± 0.0013	3.7024 ± 0.0045	3.8467 ± 0.0069
MS	2.8134 ± 0.2009	3.1003 ± 0.1541	3.3481 ± 0.1215	3.5668 ± 0.0874	3.7389 ± 0.0653	3.8785 ± 0.0520
MSG	2.7767 ± 0.1979	3.0699 ± 0.1605	3.3224 ± 0.1387	3.5375 ± 0.1134	3.7147 ± 0.0997	3.8639 ± 0.0896
HQ	2.4722 ± 0.0439	2.8140 ± 0.0401	3.1105 ± 0.0423	3.3610 ± 0.0440	3.5676 ± 0.0425	3.7392 ± 0.0407

Table 10
East OOD recursive-rollout performance over extended lead buckets.

Model	$t+1$ – $t+6$				$t+7$ – $t+12$				$t+13$ – $t+24$				$t+25$ – $t+36$			
	RMSE	MAE	R^2		RMSE	MAE	R^2		RMSE	MAE	R^2		RMSE	MAE	R^2	
N-HITS	4.5809 ± 1.0438	2.5411 ± 0.7035	0.6418 ± 0.1206		6.4437 ± 1.6762	3.6971 ± 1.1217	0.2579 ± 0.2995		7.2419 ± 1.9416	4.2172 ± 1.2480	0.0101 ± 0.3958		7.2042 ± 1.5906	4.1514 ± 0.9421	0.0024 ± 0.2767	
NLinear	5.9119 ± 0.1026	3.0699 ± 0.0478	0.4965 ± 0.0087		6.6173 ± 0.2599	3.4626 ± 0.1322	0.3696 ± 0.0399		7.5549 ± 0.3566	3.9782 ± 0.1815	0.1886 ± 0.0646		8.7962 ± 0.4953	4.6413 ± 0.2522	–0.0727 ± 0.1063	
DLinear	5.8755 ± 0.7252	3.0910 ± 0.3682	0.4853 ± 0.1001		6.5494 ± 0.8724	3.4716 ± 0.4408	0.3599 ± 0.1357		7.4657 ± 1.1207	3.9860 ± 0.5583	0.1755 ± 0.2009		8.6807 ± 1.4531	4.6497 ± 0.7150	–0.0919 ± 0.3051	
BASE	3.4020 ± 0.0045	1.7738 ± 0.0139	0.7140 ± 0.0148		4.2946 ± 0.0191	2.2197 ± 0.0165	0.5890 ± 0.0055		4.6915 ± 0.0375	2.4460 ± 0.0148	0.5190 ± 0.0018		4.9976 ± 0.0844	2.6322 ± 0.0234	0.4519 ± 0.0030	
MS	3.4275 ± 0.1067	1.7004 ± 0.0884	0.7315 ± 0.0437		4.3131 ± 0.0547	2.1917 ± 0.0451	0.5945 ± 0.0049		4.6563 ± 0.0671	2.4152 ± 0.0657	0.5216 ± 0.0183		4.9074 ± 0.0514	2.5958 ± 0.0443	0.4592 ± 0.0174	
MSG	3.4014 ± 0.1278	1.7213 ± 0.0935	0.7246 ± 0.0316		4.4212 ± 0.0814	2.2801 ± 0.0444	0.5665 ± 0.0115		4.7851 ± 0.1265	2.5116 ± 0.0669	0.4909 ± 0.0189		5.0489 ± 0.1364	2.6912 ± 0.0638	0.4263 ± 0.0190	
HQ	3.2070 ± 0.0415	1.5948 ± 0.1000	0.7542 ± 0.0330		4.2412 ± 0.0627	2.1612 ± 0.0425	0.6027 ± 0.0183		4.6411 ± 0.1254	2.3937 ± 0.0247	0.5219 ± 0.0103		4.9503 ± 0.1693	2.5763 ± 0.0224	0.4542 ± 0.0161	

Table 11
Wave-height lag-sensitive diagnostics on the East OOD split.

Lead	Model	Best lag	Zero-lag corr.	Δ corr.	Direction acc.	Peak match
$t+1$	BASE	6.00 ± 0.00	0.771 ± 0.043	–0.025 ± 0.015	0.432 ± 0.006	0.496 ± 0.031
$t+1$	HQ	5.00 ± 1.73	0.846 ± 0.073	–0.052 ± 0.024	0.425 ± 0.008	0.648 ± 0.156
$t+1$	Persistence	1.00	0.991	–0.166	0.433	1.000
$t+3$	BASE	6.00 ± 0.00	0.759 ± 0.047	0.028 ± 0.006	0.453 ± 0.007	0.482 ± 0.025
$t+3$	HQ	5.67 ± 0.58	0.821 ± 0.068	0.036 ± 0.004	0.455 ± 0.004	0.621 ± 0.148
$t+3$	Persistence	3.00	0.978	0.059	0.482	1.000
$t+6$	BASE	6.00 ± 0.00	0.731 ± 0.057	0.008 ± 0.012	0.446 ± 0.007	0.456 ± 0.039
$t+6$	HQ	6.00 ± 0.00	0.773 ± 0.054	0.013 ± 0.005	0.449 ± 0.003	0.550 ± 0.103
$t+6$	Persistence	6.00	0.950	0.024	0.466	1.000

Table 12
Paired block-bootstrap confidence intervals for key East OOD comparisons.

Comparison	Metric	Delta	95% CI	p_{boot}
HQ vs. BASE	RMSE	–0.1955	[–0.2432, –0.1457]	0.002
HQ vs. BASE	MAE	–0.1795	[–0.2912, –0.0958]	0.002
HQ vs. BASE	R^2	0.0402	[0.0104, 0.0895]	0.002
HQ vs. N-HITS	RMSE	–1.3738	[–2.0452, –0.1927]	0.002
HQ vs. N-HITS	MAE	–0.9460	[–1.5193, –0.0991]	0.002
HQ vs. N-HITS	R^2	0.1124	[–0.0501, 0.2276]	0.100

5.5. Uncertainty assessment of key model differences

Table 12 reports paired moving-block bootstrap confidence intervals for the key East single-station OOD direct-forecast comparisons. The intervals support the aggregate multivariate improvement of HQ over BASE: RMSE, MAE, and R^2 deltas all exclude zero in the favorable direction. They also support the RMSE/MAE improvement of HQ over N-HITS, although the corresponding R^2 interval includes zero.

The feature-specific bootstrap results are more conservative for wave height. Although HQ has better mean wave-height RMSE, MAE, R^2 , and zero-lag correlation than BASE, the 95% confidence intervals include zero for these wave-height-only differences. This reinforces the interpretation that short-horizon wave-height forecasting in the present dataset is strongly persistence-dominated. Accordingly, we interpret the results as evidence of statistically supported aggregate multivariate improvement, with remaining limitations on highly autocorrelated wave-height dynamics rather than as a claim of uniform wave-height improvement over persistence.

5.6. Contribution of the proposed modules: ablation study

The ablation chain shows a consistent net gain after component integration. From BASE to MS, MAE improves (1.7747 → 1.7010)

Table 13
East OOD core ablation summary (BASE→HQ).

Model	RMSE ↓	MAE ↓	R^2 ↑	Δ RMSE	Δ MAE
BASE	3.4025	1.7747	0.7139	–	–
MS	3.4278	1.7010	0.7315	+0.0253	–0.0737
MSG	3.4016	1.7219	0.7246	–0.0262	+0.0209
HQ	3.2071	1.5952	0.7542	–0.1945	–0.1267

while RMSE slightly increases (3.4025 → 3.4278), indicating a manageable trade-off from multi-scale tokenization under spatial OOD. MSG recovers RMSE relative to MS (3.4016) with comparable MAE/ R^2 . This pattern is consistent with GCI acting primarily as a conservative regularizer for cross-channel coupling rather than as the main source of aggregate error reduction. The largest improvement occurs from MSG to HQ: RMSE decreases by 0.1945, MAE by 0.1267, and R^2 increases by 0.0295. These results indicate that HAQD with randomized-horizon training resolves the intermediate trade-off and contributes the largest incremental gain within the core ablation chain. The ablation therefore points to horizon adaptation as the dominant source of practical improvement, with MSPE and GCI playing supporting but still necessary roles. In other words, once a strong PatchTST representation is available, the main remaining bottleneck is not only feature extraction, but aligning the decoder and training procedure with flexible lead-time demands in operational forecasting (see Table 13).

5.7. Feature-wise behavior on hard metocean channels

Feature-wise results in this subsection are also reported on the East single-station OOD split. This subsection reports auxiliary hard-feature variants (HQ-L, HQ-R, HQ-LR) relative to the primary model HQ. For difficult channels (current speed, wave height, wind speed), HQ-R shows the highest average hard-feature R^2 (0.6013), compared with HQ (0.5766), HQ-LR (0.5709), and HQ-L (0.5550). The gain is

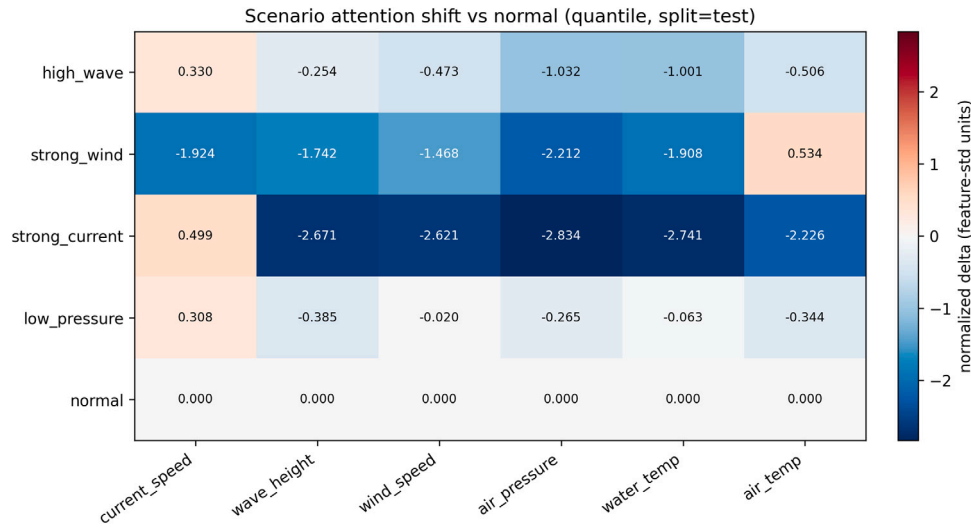


Fig. 7. Scenario-conditioned gate shifts relative to normal conditions.

Table 14

East OOD hard-feature R^2 comparison for HQ variants.

Model	Current speed	Wave height	Wind speed	Hard-feature mean
HQ	0.5668 $\pm$ 0.0195	0.6507 $\pm$ 0.1011	0.5123 $\pm$ 0.0618	0.5766
HQ-L	0.5049 $\pm$ 0.0139	0.6482 $\pm$ 0.0828	0.5120 $\pm$ 0.0533	0.5550
HQ-R	0.5536 $\pm$ 0.0285	0.6988 $\pm$ 0.0468	0.5515 $\pm$ 0.0638	0.6013
HQ-LR	0.4951 $\pm$ 0.0334	0.6815 $\pm$ 0.0786	0.5362 $\pm$ 0.0230	0.5709

concentrated on wave height and wind speed, while current-speed R^2 is slightly below that of HQ. Table 14 summarizes the channel-wise comparison. Fig. 7 reports scenario-conditioned gate-shift patterns under the strict-quantile scenario definition. The shifts are modest and mostly preserve channel-priority structure, suggesting conservative stabilization rather than strong scenario-specific feature switching. Fig. 8 additionally shows feature-wise observed-vs-predicted scatter for HQ on the test set. These results imply that hard-channel gains come less from uniformly improving all variables and more from targeted robustness on volatile channels where extremes matter most. The limited gain for current speed also suggests that some local hydrodynamic behavior may require additional spatial context or external forcing information beyond the present temporal input window.

5.8. Aggregate comparison and peak-event performance

Table 15 complements Table 5 by reporting event-focused metrics for peak-event detection. Among the learned models reported in this table, HQ achieves an F1 score of 0.9446, while its auxiliary hard-feature variant HQ-R reaches 0.9458, both exceeding the BASE configuration and the other reported learned baselines in this event-focused comparison. These results indicate that the proposed approach maintains strong performance when evaluation focuses on extreme-event detection rather than aggregate forecasting accuracy.

A clear precision–recall trade-off is observed across models. Linear baselines such as NLinear and DLinear tend to produce higher recall but lower precision, suggesting a tendency to over-flag candidate events. In contrast, MS favors precision at the expense of recall. HQ and HQ-R provide a more balanced operating point by improving recall without a comparable increase in false alarms.

From an operational perspective, this balance is desirable for event-screening systems, where excessive false positives can undermine trust in warning tools. However, threshold-based event metrics alone cannot fully capture the timing, duration, or amplitude of hazardous episodes and should therefore be interpreted as complementary indicators of warning quality. The HQ/HQ-R comparison further shows that small

Table 15

Peak-event metrics at the 90th-percentile threshold.

Model	Precision $\uparrow$	Recall $\uparrow$	F1 $\uparrow$
N-HITS	0.9185 $\pm$ 0.0366	0.8622 $\pm$ 0.0531	0.8894 $\pm$ 0.0448
NLinear	0.8992 $\pm$ 0.0028	0.9603 $\pm$ 0.0016	0.9288 $\pm$ 0.0015
DLinear	0.8903 $\pm$ 0.0042	0.9665 $\pm$ 0.0030	0.9268 $\pm$ 0.0009
BASE	0.9338 $\pm$ 0.0112	0.9438 $\pm$ 0.0099	0.9387 $\pm$ 0.0020
MS	0.9426 $\pm$ 0.0107	0.9347 $\pm$ 0.0165	0.9385 $\pm$ 0.0051
MSG	0.9209 $\pm$ 0.0161	0.9454 $\pm$ 0.0056	0.9329 $\pm$ 0.0058
HQ	0.9398 $\pm$ 0.0089	0.9495 $\pm$ 0.0110	0.9446 $\pm$ 0.0093
HQ-R	0.9404 $\pm$ 0.0106	0.9513 $\pm$ 0.0025	0.9458 $\pm$ 0.0061
HQ-L	0.9323 $\pm$ 0.0149	0.9370 $\pm$ 0.0121	0.9346 $\pm$ 0.0113
HQ-LR	0.9281 $\pm$ 0.0143	0.9469 $\pm$ 0.0113	0.9373 $\pm$ 0.0076

aggregate-error differences can meaningfully improve extreme-event ranking, supporting joint aggregate and event-focused evaluation.

6. Conclusion

We presented MetoFormer-HQ, a PatchTST-based framework for multi-step multivariate metocean forecasting that integrates multi-scale patch embedding, gated channel attention, and horizon-adaptive decoding with dynamic time queries and randomized-horizon training (Nie et al., 2023). The results suggest that the largest practical gain arises in the horizon-adaptive stage, while gated channel attention mainly serves as a conservative stabilizer of cross-variable coupling rather than a dominant source of aggregate error reduction. Overall, the proposed design addresses concurrent periodicities, noisy multivariate interactions, and flexible lead-time requirements, while showing broadly consistent learned-model behavior across East/South/West under a unified Korean coastal-buoy evaluation protocol. Paired block-bootstrap confidence intervals support the aggregate multivariate improvements of HQ over BASE on the East OOD split.

The additional iTransformer comparison shows that HQ remains strongest in East/South RMSE and MAE, but also that a recent

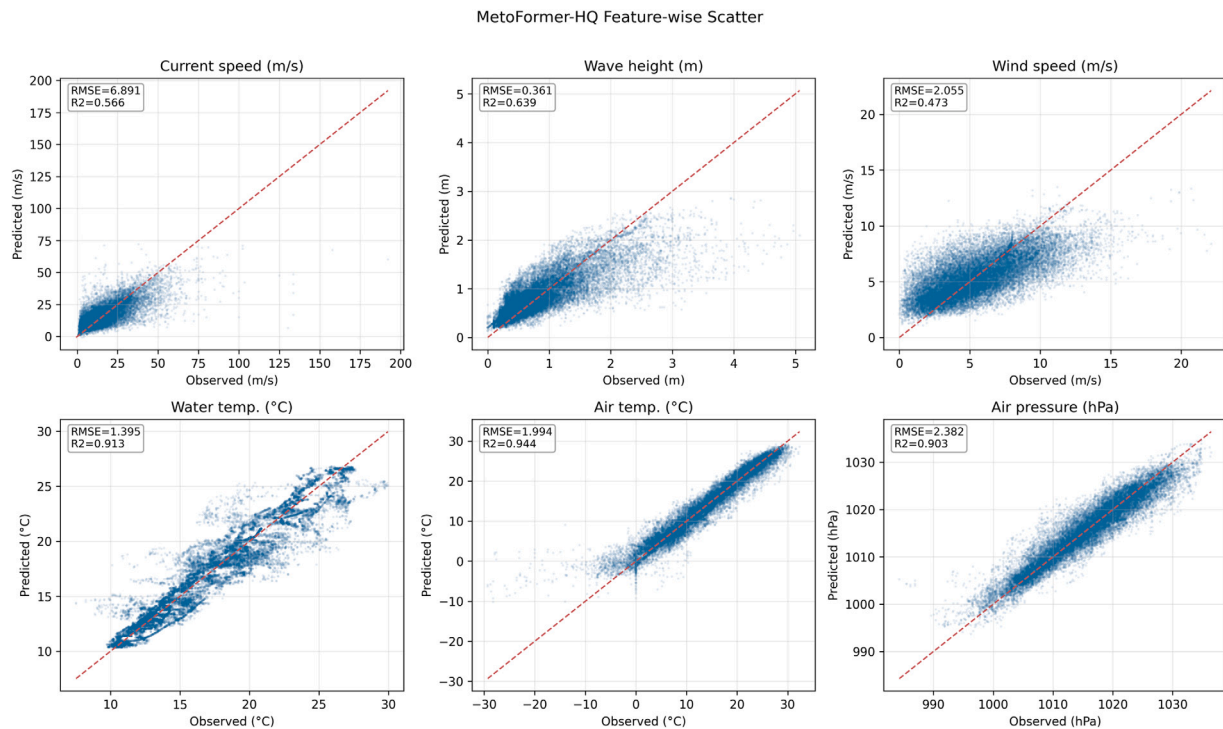


Fig. 8. Observed vs. predicted feature-wise scatter for HQ.

Transformer-family comparator performs better in the West target, leading to a region-dependent interpretation. The persistence, AR(1), lag-sensitive, and time-history diagnostics strengthen the interpretation by separating aggregate multivariate gains from the remaining difficulty of persistence-dominated wave-height prediction: HQ is not presented as uniformly better than a no-change reference on all aggregate scores, and wave-height-only confidence intervals remain inconclusive, but the analysis clarifies where the model improves aggregate forecasting and where persistence remains difficult to beat.

The study remains limited by its regional scope and by the use of station-wise temporal inputs without explicit atmospheric forcing, tide information, bathymetry, or spatial graph structure. Consequently, the results should be interpreted as evidence for Korean coastal buoy operations rather than as a globally representative validation. Future work will examine architectural choices that are currently motivated by domain knowledge and preliminary tuning, including explicit ablations on the number and composition of temporal scales. It will also extend the framework with tighter spatio-temporal modeling across larger buoy networks, richer exogenous forcings, and improved physics-guided loss formulations to better handle challenging variables such as current speed, significant wave height, and wind speed, while also capturing event timing and duration beyond threshold exceedance.

CRediT authorship contribution statement

Yunsu Kim: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **Chanjin Kwon:** Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **Hokeun Lee:** Supervision, Data curation. **Jongwon Seok:** Writing – review & editing, Supervision.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Generative AI only for language editing, sentence refinement, and improvement

of readability. The AI tool was not used to generate scientific results, interpret data, or draw research conclusions. After using this tool, the authors carefully reviewed and edited the manuscript as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The raw buoy observations used in this study are publicly accessible through the KHOA OceanGrid observation service (<http://www.khoa.go.kr/oceangrid/gis/category/observe/observeSearch.do>). The train/validation/test files used in experiments were independently curated and reprocessed by the authors from the raw observations. The processed datasets, preprocessing code, and derived experiment outputs are available from the corresponding author upon reasonable request.

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